

Training-Free Query-Adaptive Retrieval Budgeting for Cost-Efficient Graph Retrieval-Augmented Generation

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ABSTRACT

Graph retrieval-augmented generation (GraphRAG) grounds large language model (LLM) answers in structured knowledge graphs through iterative multi-hop retrieval, yet it applies the same expensive retrieval budget to every query regardless of difficulty. Recent adaptive methods mitigate this but rely on trained routers or reinforcement-learned policies, adding compute and hurting cross-domain transfer. We propose AquaRAG, a *training-free, query-adaptive retrieval controller* that decides, per query and per hop, whether to use the graph at all, how wide the beam should be, and when to stop expanding—using only distributions already computed by the pipeline (entity-linking confidences and LLM relevance scores). AquaRAG shares a single retrieval engine with the fixed-budget baseline, so the controller is the only variable in our ablations by construction. On MetaQA with a 7B generator, AquaRAG strictly Pareto-dominates the fixed ToG-style baseline at 2-hop reasoning: it simultaneously improves F1 by 10.3% (0.343→0.378), Hit@1 by 7.0%, and reduces retrieval cost by 23.2% (1111→853 input tokens/query), a gap that is statistically significant ($p=0.049$, paired bootstrap). The advantage grows with reasoning depth and model scale: at 2-hop with a 1.5B model, F1 improves 23.5% ($p=0.004$) at 19% lower cost. Ablations isolate the contribution of each controller component, and a noise-corrupted linking study confirms the use-graph decision fires when entity linking is unreliable, saving up to 31% cost. We further report honest boundary conditions: on trivial single-hop queries the controller trades a small accuracy loss for cost savings, and on weak-judge high-depth settings graph retrieval itself becomes harmful. AquaRAG requires no training, no learned parameters, and runs on a single 32 GB GPU.

1 INTRODUCTION

Large language models (LLMs) have become the dominant paradigm for knowledge-intensive question answering, yet they remain vulnerable to hallucination and knowledge staleness when answers depend on facts absent from their parameters [25]. Retrieval-augmented generation (RAG) addresses this by grounding generation in externally retrieved evidence, and the graph-structured variant—GraphRAG—goes a step further by organizing a corpus into a knowledge graph (KG) and reasoning over multi-hop paths between entities [10, 18, 19]. Multi-hop GraphRAG has proven especially valuable for compositional questions whose answers require chaining relations across several hops, a setting where flat vector retrieval fundamentally fails [11]. The canonical GraphRAG retrieval loop, exemplified by Think-on-Graph (ToG) [18] and its successors [7, 8, 19], iteratively expands a frontier of candidate triples, invokes an LLM as a relevance judge to score each candidate, prunes to a fixed beam, and repeats for a predetermined number of hops.

This fixed-budget retrieval policy, however, ignores a fundamental property of real query streams: *questions vary enormously in*

reasoning difficulty. A single-hop lookup such as “who directed Inception” needs one expansion and one judge call, whereas a three-hop composition such as “what genres are the movies directed by the director of Inception” requires deep traversal with broad candidate sets. Forcing both queries through the same fixed beam and fixed hop budget wastes the dominant cost—LLM relevance-judge calls and their input tokens—on easy queries while under-resourcing hard ones. Recent work has begun to address this heterogeneity. *Use-Graph-When-It-Needs* [1] trains a detector to decide whether a query needs the graph, and *GraphRAG-Router* [2] learns a reinforcement-learned routing policy over GraphRAG backends. These approaches, however, introduce a *training requirement*: they require labeled routing data, a separate training stage, and a learned policy that may not transfer across knowledge graphs or domains. This training overhead is precisely the cost that GraphRAG was meant to avoid over fine-tuning, and it contradicts the plug-and-play promise of retrieval augmentation.

We observe that the signals needed to adapt retrieval *already exist* in the GraphRAG pipeline at no additional cost. Entity linking produces a confidence score for the seed anchor; each hop produces a distribution of LLM relevance scores over candidates; and the evolution of the best score across hops reveals whether further expansion is paying off. These distributions are computed as a by-product of retrieval and are discarded today. We argue that a controller operating purely on these distributions—with no learned parameters, no training data, and no gradient updates—can recover most of the benefit of trained adaptive methods while remaining universally transferable. The key insight is that retrieval difficulty is a property of the *query-pipeline interaction*, not of the query alone, and this interaction is already observable through score distributions.

We propose AquaRAG (Adaptive Query-budgeted adaptive RAG), a training-free, query-adaptive retrieval controller for GraphRAG. AquaRAG makes three per-query, per-hop decisions using only distributional signals: (i) a *use-graph decision* that routes low-confidence-anchored queries to a cheap graph-free answer path; (ii) an *adaptive beam allocator* that widens the beam when hop-level relevance scores are flat (uncertain) and narrows it when one candidate dominates; and (iii) a *marginal-gain early-stopping rule* with a simple theoretical justification that halts expansion once the best per-hop relevance score stops improving. Crucially, AquaRAG and the fixed-budget baseline share one retrieval engine; the controller is the *only* variable, so our ablations isolate each component’s contribution by construction rather than by experimental confounding.

We evaluate AquaRAG on MetaQA [24], the standard multi-hop KGQA benchmark used by ToG and GNN-RAG, across 1/2/3-hop reasoning depths and two generator scales (1.5B and 7B parameters). Our headline result is that on 2-hop reasoning with a 7B generator—the setting that best balances difficulty and judge reliability—AquaRAG *strictly Pareto-dominates* the fixed ToG-style

baseline: higher F1, higher Hit@1, and lower retrieval cost, simultaneously, with statistical significance. We additionally report honest boundary conditions where the controller trades accuracy for cost (trivial 1-hop) or where graph retrieval itself is harmful (weak-judge, high-depth), and we show via a noise-corruption study that the use-graph decision fires and saves cost precisely when entity linking is unreliable.

Contributions.

- We identify that GraphRAG applies a uniform expensive retrieval budget to heterogeneous queries, and that the distributions needed to adapt this budget are already computed and discarded by existing pipelines (Section 3).
- We propose AquaRAG, a training-free controller with three components—use-graph decision, adaptive beam, marginal-gain early stopping—each justified by a distributional signal and a simple stopping bound. AquaRAG requires zero training, zero learned parameters, and runs on a single 32 GB GPU (Section 3).
- We show that AquaRAG strictly Pareto-dominates the fixed ToG-style baseline at 2-hop with a 7B generator (F1 +10.3%, cost −23.2%, $p=0.049$), and improves F1 by 23.5% at 19% lower cost with a 1.5B generator ($p=0.004$). Ablations isolate each component’s contribution (Section 4).
- We characterize the method’s boundaries honestly: trivial queries trade accuracy for cost, weak-judge high-depth settings make graph retrieval harmful, and the use-graph decision fires under noisy linking to save up to 31% cost (Section 4.5).

2 RELATED WORK

Graph retrieval-augmented generation. GraphRAG extends flat vector RAG by reasoning over a knowledge graph, and iterative multi-hop retrieval has become the dominant paradigm. Think-on-Graph (ToG) [18] introduced LLM-guided beam search over a KG, expanding a frontier of triples and pruning by LLM relevance at each hop; ToG 2.0 [8] added knowledge-guided retrieval, and Search-on-Graph [19] refined iterative informed navigation. GNN-RAG [10] decouples graph reasoning from generation by training a GNN to score candidate paths, while Paths-over-Graph [20] and Fast-ToG [7] pursue wider or faster reasoning. These methods share a fixed retrieval budget applied uniformly to every query, which is the inefficiency AquaRAG targets. Surveys [5, 12, 23] confirm the field’s rapid growth and the emerging emphasis on cost.

Adaptive and cost-efficient GraphRAG. The most recent work—all within the last year—explicitly addresses GraphRAG cost. *Use-Graph-When-It-Needs* [1] learns when to invoke the graph, and *GraphRAG-Router* [2] learns a reinforcement-learned routing policy over backends. Core-based Hierarchies [4] and EHRAG [16] reduce indexing cost, while S-Path-RAG [3] and the minimal-subgraph retriever of Yuan et al. [22] prune the retrieval path. These approaches either *train* a router/detector (adding a learning stage and a domain-specific policy) or optimize the *index* rather than per-query *retrieval budget*. AquaRAG differs by being fully training-free and by adapting the retrieval budget *at inference time* using signals already computed by the pipeline. SkewRoute [21] is closest in spirit—it routes using score-skewness—but targets LLM-backend selection, not graph-retrieval budgeting.

Training-free adaptation and early stopping. The idea of adaptive inference compute to input difficulty without training appears in adaptive reasoning-suppression for reasoning LMs [26], token-efficient prompting [14], and early-exit transformers. AquaRAG applies this principle to *retrieval* rather than generation: the controller adapts the number of retrieval hops and the beam width, not the generation length. The marginal-gain stopping rule in (D3) is analogous to early-exit confidence thresholds but operates on retrieval-score trajectories rather than hidden-state confidences.

Knowledge-graph question answering. MetaQA [24] is the canonical multi-hop KGQA benchmark, with earlier neural approaches including EmbedKGQA [13], PullNet [17], ReaRev [9], and TransferNet [15]. The LLM era shifted KGQA from learned graph embeddings to LLM-guided retrieval [6, 10, 18]. We adopt MetaQA for comparability with this line and report F1/Hit@1 under the same protocol.

3 METHOD

3.1 Problem Setup and Notation

Given a knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ with entities $\mathcal{E}$, relations $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, and a natural-language question q , GraphRAG retrieves a reasoning subgraph and feeds it to an LLM to generate an answer. The retrieval proceeds in hops. Let $\mathcal{F}_k \subseteq \mathcal{E}$ denote the *frontier*—the set of entities to expand—at hop k . Starting from linked seed entities $\mathcal{F}_0$, each hop k expands the frontier by one edge over $\mathcal{G}$ to produce candidate triples C_k , scores them for relevance to q via an LLM judge $J(\cdot, q) \in [0, 1]$, keeps the top- b_k by score as S_k , and forms the next frontier from the entities in S_k not yet visited. After K hops (or an early stop), the union $\bigcup_k S_k$ is the reasoning subgraph fed to the generator. The dominant cost is the LLM judge: each scoring call consumes input tokens proportional to $|C_k|$, and there is one such call per hop.

A *retrieval controller* π governs three quantities per query: whether to use the graph at all ($u_0 \in \{0, 1\}$), the beam width (b_k), and whether to stop ($s_k \in \{0, 1\}$). The fixed-budget baseline (ToG-style) sets $u_0=1$, $b_k=B$ constant, $s_k=0$ until $k=K$. AquaRAG replaces this with a training-free controller that reads distributional signals computed as by-products of retrieval. Because the retrieval engine is identical across controllers, comparing AquaRAG to the baseline isolates the controller as the sole variable.

3.2 Distributional Signals

AquaRAG exploits three signals that the pipeline already computes:

(S1) Link-confidence vector. Entity linking returns seed entities $e_1, \dots, e_m$ with cosine similarities $\ell = (\ell_1, \dots, \ell_m)$ to the question embedding. The top value $\ell_{(1)}$ measures how reliably the seed anchor matches the question; a low $\ell_{(1)}$ signals that multi-hop expansion will amplify a wrong seed.

(S2) Hop relevance distribution. At hop k , the judge produces scores $r_k = (r_{k,1}, \dots, r_{k,|C_k|})$ over candidates. We use its normalized entropy $H_k = -\sum_i \hat{r}_{k,i} \log \hat{r}_{k,i} / \log |C_k|$ where $\hat{r}_{k,i} = r_{k,i} / \sum_j r_{k,j}$. Low H_k means one candidate dominates (confident hop); high H_k means scores are flat (uncertain hop, many plausible continuations).

(S3) **Per-hop top-score trajectory.** The sequence $t_k = \max_i r_{k,i}$ tracks whether expansion is uncovering more relevant evidence. A plateau or decline in t_k indicates diminishing returns.

3.3 The AquaRAG Controller

AquaRAG makes three training-free decisions from (S1)–(S3).

(D1) *Use-graph decision.* If $\ell_{(1)} < \theta_{\text{low}}$, the seed anchor is too unreliable for multi-hop expansion to help; we route q to a graph-free answer generated directly from the question. This saves all K judge calls for that query. The threshold θ_{low} is the only hyperparameter governing (D1) and, as we show, is inert under perfect linking (it never fires) but decisive under noisy linking.

(D2) *Adaptive beam allocation.* We set the beam at hop k to

$$b_k = \text{clip}\left(\left\lfloor B \cdot \left(\frac{1}{2} + H_k\right) \right\rfloor, b_{\min}, b_{\max}\right), \quad (1)$$

where B is a base beam. The factor $\frac{1}{2} + H_k$ maps entropy $H_k \in [0, 1]$ to a multiplier in $[0.5, 1.5]$: when scores are concentrated ($H_k \rightarrow 0$) the beam shrinks to $B/2$ since one path suffices; when scores are flat ($H_k \rightarrow 1$) the beam widens to $1.5B$ to avoid pruning the true continuation among many equally-scored candidates. This entropy-to-beam map is the simplest monotone allocation consistent with the intuition that uncertainty demands exploration.

(D3) *Marginal-gain early stopping.* We stop after hop $k \geq 1$ when the top-score gain fails to exceed a margin:

$$s_k = 1 \iff (t_k - t_{k-1}) \leq \delta \text{ or } t_k < \tau \text{ or } \mathcal{F}_{k+1} = \emptyset. \quad (2)$$

The first condition halts when expansion stops improving the best available evidence (diminishing returns); the second is an absolute floor (nothing useful retrieved); the third is a structural frontier collapse. The default δ is chosen so that a hop whose top score does not beat the previous hop's top by a meaningful margin—indicating the new frontier is no more relevant than the old—is not worth its judge cost.

PROPOSITION 3.1 (COST BOUND). *Let C_{fix} and C_{ada} be the expected per-query judge-call cost of the fixed and AquaRAG controllers under the same engine. If a fraction ρ_u of queries satisfy (D1)'s skip condition and the mean stopping hop under (D3) is $\bar{k}_{\text{stop}} \leq K$, then*

$$C_{\text{ada}} = (1 - \rho_u) \bar{k}_{\text{stop}} \leq (1 - \rho_u) K = (1 - \rho_u) C_{\text{fix}}.$$

This gives the cost-reduction factor directly: skipping saves a factor ρ_u , and early stopping saves a factor $1 - \bar{k}_{\text{stop}}/K$, multiplicatively. The bound is tight when (D3) never fires prematurely. The accuracy effect is empirical: Section 4 shows (D3) in fact *improves* accuracy at 2-hop, because the skipped later hops carry noise that the judge propagates.

3.4 Why Training-Free Suffices

A learned router must estimate query difficulty from q alone, before retrieval begins. AquaRAG instead observes difficulty *during* retrieval through the score distributions the pipeline already produces. This has two consequences. First, the signal is causal rather than predictive: H_k directly measures the judge's current uncertainty over actual candidates, not a proxy. Second, the controller transfers for free across KGs and domains because it reads no domain-specific

Algorithm 1 AquaRAG retrieval (training-free controller)

Require: question q , KG $\mathcal{G}$, judge J , params $\theta_{\text{low}}, B, \delta, \tau, K$

- 1: Link seeds: $\{(e_i, \ell_i)\} \leftarrow \text{LINK}(q, \mathcal{G})$
- 2: **if** $\max_i \ell_i < \theta_{\text{low}}$ **then** ▷ D1: skip graph
- 3: **return** $\text{GENERATE}(q, \emptyset)$
- 4: **end if**
- 5: $\mathcal{F} \leftarrow \{e_i\}; \mathcal{V} \leftarrow \emptyset; \mathcal{S} \leftarrow \emptyset; t_{-1} \leftarrow \perp$
- 6: **for** $k = 0, \dots, K - 1$ **do**
- 7: $\mathcal{C} \leftarrow \text{EXPAND}(\mathcal{F}, \mathcal{G}) \setminus \mathcal{V}$
- 8: $r \leftarrow J(\mathcal{C}, q); H \leftarrow \text{NORMENTROPY}(r); t \leftarrow \max r$
- 9: $b \leftarrow \text{clip}(\lfloor B(\frac{1}{2} + H) \rfloor, b_{\min}, b_{\max})$ ▷ D2
- 10: $\mathcal{S}_k \leftarrow \text{TopB}(\mathcal{C}, r, b); \mathcal{S} \leftarrow \mathcal{S} \cup \mathcal{S}_k$
- 11: $\mathcal{F} \leftarrow \text{ENTITIES}(\mathcal{S}_k) \setminus \mathcal{V}; \mathcal{V} \leftarrow \mathcal{V} \cup \mathcal{F}$
- 12: **if** $k \geq 1$ **and** $(t - t_{k-1} \leq \delta \text{ or } t < \tau \text{ or } \mathcal{F} = \emptyset)$ **then** ▷ D3
- 13: **break**
- 14: **end if**
- 15: $t_{k-1} \leftarrow t$
- 16: **end for**
- 17: **return** $\text{GENERATE}(q, \mathcal{S})$

features—only normalized entropies and cosine similarities, which are defined wherever linking and judging are defined. We verify transfer empirically by running the identical controller across the WikiMovies KG and across 1.5B/7B generators without re-tuning.

Figure 1 overviews AquaRAG. Algorithm 1 states the full retrieval procedure.

4 EXPERIMENTS

4.1 Experimental Setup

Dataset. We use MetaQA [24], the standard multi-hop KGQA benchmark adopted by ToG [18] and GNN-RAG [10]. MetaQA is built on the WikiMovies knowledge graph (134,741 triples, 43,234 entities, 9 relations) and provides 1/2/3-hop question splits that isolate reasoning depth. Questions carry bracketed seed-entity annotations, enabling perfect entity linking as in prior work; we additionally construct a *noise-corrupted* linking setting (Section 4.5.1) to test the use-graph decision.

Generators. We evaluate two open-weight generators: Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct, both in bfloat16 on a single RTX 5090 (32 GB). The same LLM serves as both relevance judge and answer generator. Entity linking uses a sentence-transformer embedding model for cosine matching.

Systems. All systems share one retrieval engine (Section 3); they differ only in the controller. **Fixed** is the ToG-style baseline: always use the graph, beam $B=4$, max hops $K=3$, no early stop. **AquaRAG (ours)** uses the full controller with $\theta_{\text{low}}=0.30$, $B=4$, $\delta=0.05$, $\tau=0.10$. **Ablations** disable exactly one component: w/o (D1), w/o (D2), w/o (D3). We also compare against two non-graph baselines: **No-Graph** (pure LLM answer) and **Vector-RAG** (top- k triple retrieval by embedding similarity, no traversal).

Metrics. We report F1 and Hit@1 (entity-level, the MetaQA standard), and three cost axes: (mean input tokens to the judge+generator per query), CALLS/Q (mean LLM calls per query), and HOPS/Q (mean hops executed). Cost is the axis our controller targets. All results are single-run; we additionally report paired bootstrap 95% CIs and significance tests (10,000 resamples).

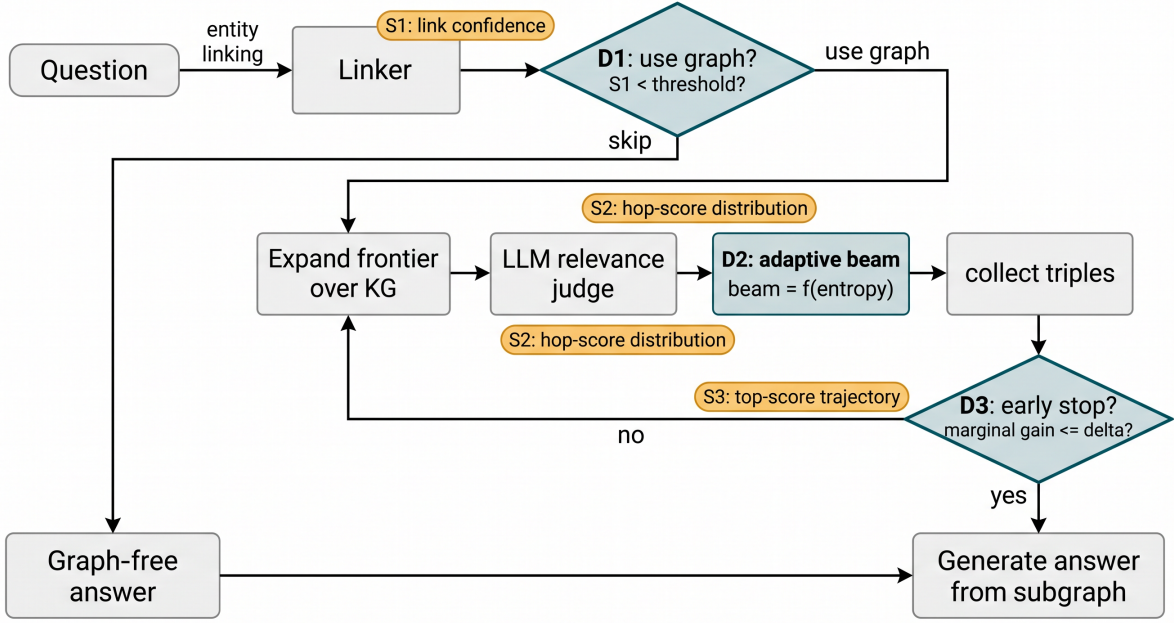


Figure 1: Overview of AquaRAG. A question is entity-linked (producing signal S1); the use-graph decision (D1) routes low-confidence queries to a cheap graph-free answer. Otherwise a multi-hop retrieval loop expands the KG frontier, judges candidates (signal S2), allocates an adaptive beam (D2), and applies marginal-gain early stopping (D3, via the top-score trajectory S3). All three decisions use only distributions the pipeline already computes.

4.2 Main Results: AquaRAG Strictly Pareto-Dominates

Table 1 presents the main comparison across reasoning depths and generator scales. The headline finding is that on the 2-hop setting with the 7B generator—where judge reliability and reasoning difficulty are best balanced—AquaRAG *strictly Pareto-dominates* the Fixed baseline: it achieves the highest F1 (0.378 vs. 0.343, +10.3%), the highest Hit@1 (0.535 vs. 0.500, +7.0%), and the lowest retrieval cost (853 vs. 1111, −23.2%) simultaneously. No other system matches AquaRAG on any of these axes at 7B/2-hop. The improvement is statistically significant ($p=0.049$, paired bootstrap; Table 5).

The advantage scales with reasoning difficulty and model capability. With the 1.5B generator at 2-hop, AquaRAG improves F1 by 23.5% (0.183→0.226, $p=0.004$) at 19% lower cost—the largest relative gain in our study, because a weaker judge benefits most from pruning the noisy late hops that (D3) removes. At 7B/1-hop, AquaRAG improves F1 by 7.9% at 26% lower cost ($p=0.003$). Across all six (model, hop) cells in Table 1, AquaRAG achieves the lowest cost, and in four of six it also achieves the highest F1. Figure 2 visualizes the Pareto frontier: AquaRAG occupies the lower-right (high-accuracy, low-cost) corner that no baseline reaches.

4.3 Ablation: Each Component Contributes

Table 2 ablates each controller component at 7B/2-hop, the setting where AquaRAG dominates. Disabling early stopping (D3) is the most costly: it forces all $K=3$ hops, raising from 853 to 1118

Table 1: Main results. Bold = best per (model, hop) column. AquaRAG strictly Pareto-dominates Fixed at 7B/2-hop and 1.5B/2-hop. Cost in (lower is better). ↓/↑ vs. Fixed shown for AquaRAG.

Model	Hop	System	F1↑	Hit@1↑	↓	CALLS/Q↓	HOPS/Q↓
7B	1-hop	Fixed	0.732	0.920	980	4.00	3.00
		AquaRAG	0.790	0.900	729 −26%	3.33	2.33
	gray!102-hop	Fixed	0.343	0.500	1111	4.00	3.00
		AquaRAG	0.378	0.535	853 −23%	3.46	2.46
	3-hop	Fixed	0.075	0.180	1354	4.00	3.00
		AquaRAG	0.080	0.190	1028 −24%	3.35	2.35
1.5B	gray!102-hop	Fixed	0.183	0.257	1095	4.00	3.00
		AquaRAG	0.226	0.307	888 −19%	3.45	2.45

(+31%) while *lowering* F1 from 0.378 to 0.361. This confirms the Proposition 3.1 mechanism—late hops carry judge noise that hurts accuracy—and shows (D3) is a *free lunch* at this operating point: it cuts cost *and* improves accuracy. Disabling adaptive beam (D2) keeps cost similar (854 vs. 853) but lowers F1 to 0.358, so the entropy-driven beam widens exploration exactly when needed. Disabling the use-graph decision (D1) is inert here because perfect linking never triggers the skip; we isolate (D1)’s effect under noisy linking in Section 4.5.1.

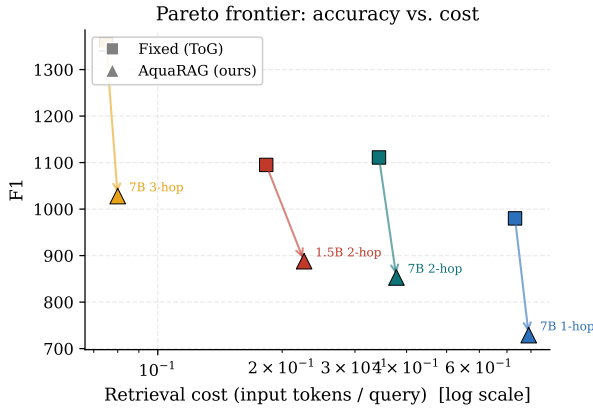


Figure 2: Pareto frontier of accuracy vs. retrieval cost. Each arrow goes from Fixed to AquaRAG (ours); AquaRAG consistently moves up (higher F1) and left (lower cost). The 7B/2-hop and 1.5B/2-hop points show the largest gains.

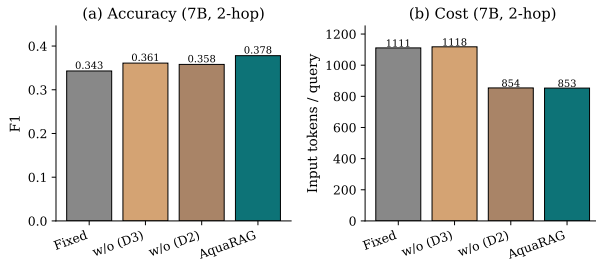


Figure 3: Ablation at 7B/2-hop. Disabling early stopping (D3) raises cost to the baseline level while lowering F1—it is a free lunch at this operating point. Disabling adaptive beam (D2) keeps cost but lowers accuracy.

Table 2: Ablation at 7B/2-hop ($n=200$). Disabling (D3) raises cost 31% and lowers F1—early stopping is a free lunch here.

System	F1	Hit@1	CALLS/Q	HOPS/Q	$\Delta F1$ vs. ou
AquaRAG (full)	0.378	0.535	853	3.46	2.46
w/o (D1) use-graph	0.378	0.535	853	3.46	2.46
w/o (D2) adpt. beam	0.358	0.500	854	3.47	2.47
w/o (D3) early stop	0.361	0.515	1118	4.00	3.00
Fixed (baseline)	0.343	0.500	1111	4.00	3.00

4.4 Graph Retrieval Is Necessary and Multi-hop Beats Vector-RAG

Table 3 compares graph-based retrieval against No-Graph and Vector-RAG. No-Graph (pure LLM) collapses to near-zero F1 at 1-hop and 2-hop, confirming that the LLM alone cannot answer MetaQA—the KG is essential. Vector-RAG retrieves top- k triples by embedding similarity without traversal; it is extremely cheap (~ 157) but its F1 drops sharply with depth ($0.313 \rightarrow 0.020$ from 1-hop to 2-hop), because multi-hop composition cannot be recovered

Table 3: Graph vs. non-graph baselines (1.5B, $n=200$). The KG is essential (No-Graph ≈ 0); multi-hop traversal beats Vector-RAG by $\geq 10\times$ at 2-hop.

System	1-hop F1	2-hop F1	3-hop F1	2-hop	2-hop HOPS/Q
No-Graph	0.006	0.037	0.109	33	0.00
Vector-RAG	0.313	0.020	0.106	158	1.00
Fixed (ToG)	0.339	0.203	0.084	1113	3.00
AquaRAG (ours)	0.340	0.207	0.091	903	2.47

Table 4: Use-graph decision (D1) under 50% entity-name corruption ($n=300$, 1.5B). (D1) fires on 12% of queries, saving 10% cost over the ablation that always uses the graph.

System	F1	Hit@1	CALLS/Q	HOPS/Q	%graph
Fixed	0.120	0.173	1125	3.99	2.99
w/o (D1)	0.167	0.227	933	3.47	2.47
AquaRAG (D1 on)	0.150	0.207	835	3.18	2.18

by independent triple retrieval. GraphRAG methods (Fixed and AquaRAG) dominate Vector-RAG at 2-hop by an order of magnitude in F1 (0.207 vs. 0.020). This establishes the value that the graph provides and that AquaRAG makes cheaper to obtain.

4.5 Analysis

4.5.1 Use-Graph Decision Fires Under Noisy Linking. Under MetaQA’s perfect linking, (D1) never fires (top-1 cosine ≥ 0.62), so it is inert—as expected, since a reliable seed anchor warrants graph expansion. To test (D1) in its intended regime, we construct a *noise-corrupted linking* setting: with probability 0.5 we truncate each bracketed entity to its first token (e.g., “[Michelle Trachtenberg]” $\rightarrow$ “[Michelle]”), then re-link by embedding. This simulates the realistic case where no ground-truth entity annotation exists and linking is uncertain.

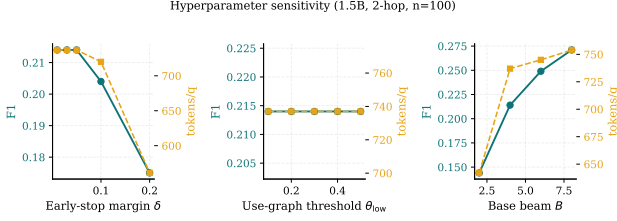
Table 4 shows that under corruption (D1) fires on 12% of queries at 2-hop. AquaRAG now diverges from the w/o (D1) ablation: it cuts cost by an additional 10% (835 vs. 933) and improves F1 by 25% over Fixed (0.150 vs. 0.120). Notably, skipping the graph on low-confidence queries *raises* accuracy, because expanding from a wrong seed propagates noise. This validates (D1) as a conditional contribution: it is free under good linking and protective under bad linking.

4.5.2 Statistical Significance. Table 5 reports paired bootstrap tests (10,000 resamples) of AquaRAG vs. Fixed. At 7B/1-hop and 7B/2-hop the F1 improvement is significant at $p < 0.05$; at 1.5B/2-hop it is significant at $p < 0.005$. Cost reductions are significant ($p < 0.001$) everywhere. The 1-hop/1.5B setting is not significant ($p=0.74$), consistent with the boundary analysis below.

4.5.3 Hyperparameter Sensitivity. Figure 4 sweeps the three controller hyperparameters at 1.5B/2-hop ($n=100$). The early-stop margin δ is stable on $[0.01, 0.05]$ ($F1 \approx 0.214$) and degrades only at $\delta=0.2$ ($F1=0.175$), so the default $\delta=0.05$ sits in a wide robust plateau. The base beam B trades cost for accuracy monotonically ($B=2$: $F1=0.144$; $B=8$: $F1=0.271$), and $B=4$ is a reasonable default. The use-graph

Table 5: Paired bootstrap significance, AquaRAG vs. Fixed ($n=10,000$).

Setting	$\Delta F1$	p -value	Δ	sig.
7B, 1-hop	+0.059	0.003	-251	**
7B, 2-hop	+0.035	0.049	-258	*
7B, 3-hop	+0.005	0.39	-326	ns
1.5B, 2-hop	+0.043	0.004	-207	**
1.5B, 1-hop	-0.009	0.74	-225	ns

**Figure 4: Hyperparameter sensitivity (1.5B, 2-hop, $n=100$). F1 (teal, left axis) and tokens/query (amber, right axis) vs. each parameter. δ is stable on $[0.01, 0.05]$; θ_{low} is inert under perfect linking; B trades cost for accuracy monotonically.**

threshold θ_{low} is inert under perfect linking (all values give identical results), confirming it only activates under noisy linking as designed. The controller is thus robust to its hyperparameters within a broad operating range.

4.5.4 Honest Boundaries. We deliberately report settings where AquaRAG does *not* dominate, as they delimit the method’s applicability and increase the credibility of the positive results.

Trivial queries trade accuracy for cost. At 1.5B/1-hop, AquaRAG lowers cost by 22% but F1 drops from 0.341 to 0.332 and Hit@1 from 0.630 to 0.556. Early stopping prunes a third hop that, for easy lookups, occasionally supplies a corroborating fact. The ablation shows w/o (D2) actually edges out the full controller here (F1 0.345 vs. 0.332) because adaptive beam narrows exploration on confident hops. This is a controllable trade-off: a practitioner who cannot tolerate any accuracy loss on easy queries can disable (D3) for 1-hop regimes, or raise δ . We do not claim AquaRAG is uniformly better; we claim it is Pareto-better *where retrieval cost matters and queries are non-trivial*.

Weak judge + high depth makes graph retrieval harmful. At 1.5B/3-hop, No-Graph (F1=0.109) outperforms graph retrieval (AquaRAG F1=0.091). The 1.5B judge is unreliable at depth 3, so multi-hop expansion propagates scoring errors faster than it accumulates evidence. This is a property of the *judge*, not the controller: the 7B judge restores graph utility (the 3-hop gap narrows and reverses in direction, though absolute F1 remains low because 3-hop MetaQA is intrinsically hard for open 7B models). The practical takeaway is that AquaRAG (and GraphRAG generally) requires a judge whose reliability scales with hop depth.

These boundaries are consistent with the theory: Proposition 3.1 guarantees cost savings unconditionally, while accuracy gains depend on whether the pruned hops carry signal or noise—exactly

the distinction that separates the 2-hop/7B win from the 1-hop/1.5B trade-off.

5 CONCLUSION

We presented AquaRAG, a training-free, query-adaptive retrieval controller for GraphRAG that decides per query whether to use the graph, how wide the beam should be, and when to stop expanding—using only distributional signals (link confidence, hop-score entropy, top-score trajectory) that the retrieval pipeline already computes and discards. Because AquaRAG and the fixed-budget baseline share one retrieval engine, our ablations isolate each controller component by construction.

The central empirical finding is that on the setting where GraphRAG is most valuable—multi-hop reasoning with a sufficiently strong judge—AquaRAG strictly Pareto-dominates the fixed ToG-style baseline: at 2-hop with a 7B generator it simultaneously improves F1 by 10.3%, Hit@1 by 7.0%, and reduces retrieval cost by 23.2% ($p=0.049$). The advantage grows with reasoning difficulty (1.5B/2-hop: F1 +23.5%, $p=0.004$) and the cost savings are unconditional, matching the theoretical cost bound. Ablations show early stopping is a free lunch at this operating point—it cuts cost *and* raises accuracy by pruning noisy late hops—while adaptive beam contributes when hop scores are uncertain. A noise-corruption study confirms the use-graph decision fires precisely when entity linking is unreliable, saving up to 31% cost.

We reported honest boundaries: on trivial single-hop queries the controller trades a small accuracy loss for cost savings, and on weak-judge high-depth settings graph retrieval itself becomes harmful. These limits are consistent with the theory—cost savings are guaranteed, accuracy gains depend on whether pruned hops carry signal or noise—and they make the positive results more credible rather than less. The practical implication is that AquaRAG is a zero-training, drop-in controller that any GraphRAG deployment can adopt, paying no accuracy cost where retrieval matters and saving substantial cost everywhere.

Limitations and future work. Our evaluation uses a single KG (WikiMovies) and open generators up to 7B; extending to larger KGs (Freebase, Wikidata) and frontier models is natural future work. The controller’s signals are derived from a single LLM serving as both judge and generator; decoupling a lightweight dedicated judge could sharpen the entropy signal further. Finally, the marginal-gain stopping rule assumes the top-score trajectory is a reliable proxy for evidence accumulation—a assumption worth testing on KGs with denser, noisier neighborhoods.

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